

Introduction to Arduino Computing and Electronics

This laboratory-based course will introduce students to the world of computer programming and electronics. Students who are fully engaged in this course (which means completing the homework assignments in addition to in-class work) will be well prepared for future computer science courses in high school. They will also develop skills in creative problem solving and troubleshooting. Homework problems early in the year focus on designing simple circuits with LEDs and making them blink in designed patterns. The final project in the course is to design and build a working 4 digit LED-Display clock that accurately keeps time. This course is a great foundation for CT Science Fair and Invention Convention projects.

We will use the Arduino platform, which was specifically designed for education, and we will be programming in the C++ language. The Arduino is an inexpensive computer that excels in controlling electronic circuits. Arduino boards are able to read inputs (such as light on a sensor, temperature, a finger on a button or even a Twitter message) and turn it into an output (activating a motor or relay, turning on an LED, publishing an online message). The platform has been used extensively for monitoring sensors, robotics, home automation and other applications. Students will learn basic electronics, including how to construct a circuit on a breadboard, how to read a schematic diagram, the role of different electronic components in a circuit and how to design simple circuits. Students will also learn computer programming using the Arduino Integrated Development Environment. Concepts to be covered include reading and constructing a computer flowchart, understanding and debugging code, learning how to use a compiler and principles of programming in C++. There are no course prerequisites other than a genuine interest in science. Students must purchase the Elegoo Mega 2560 Project Starter Kit (EL-KIT-008 on Amazon - \$70). Students must also have their own laptop computer running the Arduino IDE (available for download free at www.arduino.cc/en/Main/Software) with an available USB port. Due to the presence of numerous small parts in the kit, this course is only available for students in Grade 6 or above. High school students and adults are also welcome. Enrollment will be limited to 20 students.